

**NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**EXAMINATION: End Semester**  
**SUBJECT: Computer Vision**

■ CS 6220 ■ Duration 120 minutes ■ Full Marks 50 ■ Date: 11-December-2021■

---

*After Completing the answer you need to mail me the digital copy of the answer sheet.*  
*Number within bracket indicate full mark for that question.*

---

1. (a) Prove that the eigen vectors of a real, symmetric matrix with distinct eigenvalues are orthogonal. [4]  
(b) Compute the singular value decomposition. [4]

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

2. (a) Construct a fully populated approximation pyramid and corresponding prediction residual pyramid for the following image [4]

$$f(m, n) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix}$$

- (b) Explain the Basic operations used a CNN. How it is different than ANN? [4]
3. (a) If two pattern classes are identified by their mean vectors [4]

$$\mu_1 = [1.2 \quad 1]^T$$
$$\mu_2 = [2 \quad 2.3]^T$$

Find the equation of the boundaries for a minimum distance classifier.

- (b) What is sub-band coding? How it is helpful in Image analysis? Explain with an example. [4]
4. (a) How many types of orientations are possible in calibration? Illustrate the difference among them with some examples. [4]  
(b) What are the intrinsic and extrinsic calibration parameters of a single-perspective camera? How are they estimated from known scenes? [4]
5. (a) Give examples of perspective images from everyday life. Why do parallel lines in the world not correspond to parallel ones in images? [4]  
(b) Will the same unit square be preserved if the camera coordinate system captures a unit square in the world coordinate system? Answer with adequate examples and explanations. [4]
6. Write short note on any of the following two. [5+5]
- (a) Human gait analysis  
(b) Locating roadway  
(c) Face recognition

\*\*\*\*\*Wish you all the best\*\*\*\*\*